THE PLANAR XOR TWO-NEIGHBOUR GRAPH IS NOT SELF-DUAL: EXACT COMPUTATIONS AND EVIDENCE FOR SUBCRITICALITY

RAW AI OUTPUT (CLAUDE OPUS 5.5), PREPARED AT THE REQUEST OF J. KÖPPL

ABSTRACT. In the 2-XnG on $\mathbb{Z}^2$ every vertex chooses two of its four neighbours uniformly at random, and a nearest-neighbour edge is kept if exactly one of its endpoints chose it. Every edge is open with probability $1/2$, and the closed edges have the same law as the open ones. This suggests that the model is self-dual and critical, which would give absence of percolation together with a lower bound $\mathbb{P}(o \leftrightarrow \partial B_n) \geq c/n$ on the one-arm probability.

We show that the model is *not* self-dual. The planar dual does not have the law of the primal, and the probability that $[0, n+1] \times [0, n]$ is crossed horizontally is strictly smaller than $1/2$ for $n = 2, 3$, where we compute it exactly. Along the way we compute the law of the model in terms of characters and show that the Harris–FKG inequality fails. We also place the 2-XnG in a family of "vertex-XOR" models that contains critical bond percolation and (strictly subcritical) AB percolation. To first order, the 2-XnG is a small perturbation of critical bond percolation in the direction of AB percolation.

Monte Carlo simulations up to $n = 4096$ show that the crossing probabilities keep decreasing, to about 0.30 at $n = 4096$. The deficit grows like $n^{3/4}$, which is the off-critical scaling of planar percolation. This strongly supports the conjecture of [JKLT25] that the 2-XnG does not percolate in $d = 2$. It also indicates that the model is slightly *subcritical*. If so, the one-arm probability decays exponentially beyond a correlation length of order 10^3–10^4, and the requested polynomial lower bound is false. We have no rigorous proof of either statement; we explain where the standard arguments fail and record what is proved.

1. The question and the answer in brief

Let $V = \mathbb{Z}^2$, let E be the set of nearest-neighbour edges, and let N_x be the set of the four neighbours of x. Independently for every $x \in V$, let $S_x \subseteq N_x$ be uniform among the six subsets of size two, and write $\sigma_x(y) = \mathbf{1}\{y \in S_x\}$. The *2-XnG* [JKLT25, Sec. 3.3] is the random edge configuration

$$\omega(e) = \sigma_x(y) \oplus \sigma_y(x), \qquad e = \{x, y\} \in E,$$

where $\oplus$ is addition modulo 2. An edge e is *open* if $\omega(e) = 1$. We write $\mathbb{P} = \mathbb{P}_{\mathrm{X}}$, $B_n = \{-n, \dots, n\}^2$ and $\partial B_n = B_n \setminus B_{n-1}$, and $\{o \leftrightarrow \partial B_n\}$ denotes the event that the origin is joined to ∂B_n by an open path. We were asked to show:

(A) the 2-XnG does not percolate: $\mathbb{P}(o \leftrightarrow \infty) = 0$;
(B) the one-arm probability decays at most polynomially: $\mathbb{P}(o \leftrightarrow \partial B_n) \geq c\, n^{-C}$ for some $c, C > 0$.

The natural route to both is planar self-duality, as for bond percolation at $p = 1/2$. Every edge is open with probability $1/2$, and the set of closed edges has the same law as the set of open edges (Proposition 2.3). If the dual configuration were a copy of the primal one, the rectangle $R_n = \{0, \dots, n+1\} \times \{0, \dots, n\}$ would be crossed horizontally with probability exactly $1/2$. This would give (B) in the form $\mathbb{P}(o \leftrightarrow \partial B_n) \geq 1/(2n+2)$ (Lemma 6.1), and Zhang's argument would then give (A), at least in the presence of the Harris–FKG inequality.

Findings.

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(1) *Exact law* (Proposition 2.1). $\mathbb{E}\prod_{e\in A}(-1)^{\omega(e)} = 3^{-v_2(A)}$ if A is an even subgraph and 0 otherwise, where $v_2(A)$ is the number of vertices of degree two in A. In particular, the restriction of ω to any forest is i.i.d. Bernoulli(1/2). The only deviation from critical bond percolation sits on cycles, and it is small: $3^{-4} = 1/81$ for a plaquette.
(2) *No Harris–FKG* (Proposition 2.4). There are two increasing events on a single plaquette with covariance $-1/1296$.
(3) *No self-duality* (Propositions 3.1 and 3.2). Under no isomorphism of the dual lattice onto $\mathbb{Z}^2$ does the closed dual configuration have the law of ω. Moreover, with H_n the event that R_n is crossed from left to right,

$$\mathbb{P}(H_2) = \tfrac{124345}{248832} = \tfrac{1}{2} - \tfrac{47}{165888} - \tfrac{1}{497664} < \tfrac{1}{2}, \qquad \mathbb{P}(H_3) = \tfrac{69570879977}{139314069504} \approx 0.49938 < \tfrac{1}{2}.$$

Both values are exact, and each was obtained by two independent methods (Section 3).
(4) *Where the model sits* (Section 4). Bond percolation at $p = 1/2$, AB percolation at $p = 1/2$ and the 2-XnG are the XOR graphs obtained when the number of chosen neighbours is $\mathrm{Bin}(4, 1/2)$, uniform on $\{0, 4\}$, and equal to 2, respectively. For every f depending on finitely many edges,

$$\mathbb{E}_{\mathrm{X}} f - \mathbb{E}_{\mathrm{Ber}} f = \sum_A 3^{-v_2(A)} \hat{f}(A), \qquad \mathbb{E}_{\mathrm{AB}} f - \mathbb{E}_{\mathrm{Ber}} f = \sum_A \hat{f}(A),$$

both sums running over nonempty even subgraphs (Proposition 4.2). AB percolation is strictly subcritical. For crossing events, the single-plaquette coefficients are $\hat{f}(\partial p) = -\frac{1}{16}\mathbb{P}_{\mathrm{Ber}}(E_p) \le 0$ (Proposition 4.3). To first order in this expansion, the 2-XnG is therefore critical bond percolation pushed slightly towards the subcritical side. Larger even subgraphs contribute with both signs, but for $n = 2, 3$ the single-plaquette terms account for more than 98% of the deficit.
(5) *Numerics* (Section 5).
- $\mathbb{P}(H_n)$ decreases steadily, from 0.4971 ± 0.0029 at $n = 16$ to 0.421 ± 0.011 at $n = 1024$ and 0.30 ± 0.04 at $n = 4096$. Over $32 \le n \le 1024$ the deficit $1/2 - \mathbb{P}(H_n)$ grows like $n^{0.74}$, close to the off-critical scaling $n^{3/4}$ of planar percolation.
- The one-arm probability falls below that of critical bond percolation from $n \approx 500$ on.
- Adding independent open edges with density δ restores scale invariance only at $\delta^* \approx 10^{-3}$ (Table 3).

Conclusion. Statement (A) is very probably true. In fact the model is very probably *subcritical*, and then statement (B) is false: the one-arm probability decays exponentially, with a correlation length of order 10^3–10^4, so that the decay looks polynomial on all scales accessible to simulation. Neither statement is proved here. Section 6 explains what a proof would need.

2. The law of the 2-XnG

For a finite $A \subseteq E$, write $\chi_A = \prod_{e\in A}(-1)^{\omega(e)}$. Let $\deg_A(x)$ be the degree of x in A, call A *even* if all degrees are even, and let $v_k(A) = |\{x : \deg_A(x) = k\}|$.

Proposition 2.1 (Characters)**.** *For every finite $A \subseteq E$,*

$$\mathbb{E}[\chi_A] = \begin{cases} 3^{-v_2(A)} & \text{if } A \text{ is even,} \\ 0 & \text{otherwise.} \end{cases}$$

Proof. For $e = \{x, y\}$ we have $(-1)^{\omega(e)} = (-1)^{\sigma_x(y)}(-1)^{\sigma_y(x)}$. Hence $\chi_A = \prod_{x\in V}\prod_{y\in A_x}(-1)^{\sigma_x(y)}$, where $A_x = \{y \in N_x : \{x, y\} \in A\}$ and $|A_x| = \deg_A(x)$. By independence, $\mathbb{E}[\chi_A] = \prod_x m(|A_x|)$.

Here $m(k) = \mathbb{E}(-1)^{|S\cap B|}$, where S is a uniform 2-subset of a four-element set and B is a fixed subset with $|B| = k$. Counting the six subsets gives

$$m(0) = 1, \quad m(1) = \tfrac{3-3}{6} = 0, \quad m(2) = \tfrac{1-4+1}{6} = -\tfrac{1}{3}, \quad m(3) = \tfrac{3-3}{6} = 0, \quad m(4) = 1.$$

Hence $\mathbb{E}[\chi_A] = 0$ unless A is even, and then $\mathbb{E}[\chi_A] = (-1/3)^{v_2(A)}$. It remains to see that $v_2(A)$ is even. From $2|A| = \sum_x \deg_A(x) = 2v_2(A) + 4v_4(A)$ we get $v_2(A) = |A| - 2v_4(A)$. Moreover $|A|$ is even, because an even subgraph is an edge-disjoint union of cycles, and cycles in the bipartite graph $\mathbb{Z}^2$ have even length. □

Corollary 2.2. *(a) For finite $F \subseteq E$ and $\eta \in \{0,1\}^F$,*

$$\mathbb{P}(\omega|_F = \eta) = 2^{-|F|} \sum_{A\subseteq F \text{ even}} (-1)^{\sum_{e\in A}\eta(e)}\, 3^{-v_2(A)}.$$

(b) If F contains no cycle, then $\omega|_F$ is i.i.d. Bernoulli$(1/2)$. In particular, any two or three edges are independent, the four edges at a vertex are independent, and every self-avoiding path with ℓ edges is open with probability exactly $2^{-\ell}$.

(c) If $e_1, \dots, e_4$ are the edges of a plaquette, then $\mathbb{P}(\omega(e_i) = \eta_i \ \forall i) = \frac{1}{16}\big(1 + \frac{(-1)^{\eta_1+\cdots+\eta_4}}{81}\big)$.

Proof. (a) Expand $\mathbf{1}\{\omega|_F = \eta\} = \prod_{e\in F} \frac{1}{2}\big(1 + (-1)^{\eta(e)}(-1)^{\omega(e)}\big) = 2^{-|F|}\sum_{A\subseteq F}(-1)^{\eta(A)}\chi_A$, where $\eta(A) = \sum_{e\in A}\eta(e)$, and apply Proposition 2.1. (b) A forest has no nonempty even subgraph, and a cycle of $\mathbb{Z}^2$ has at least four edges. (c) The even subgraphs of a 4-cycle are $\emptyset$ and the cycle itself, which has $v_2 = 4$. □

Proposition 2.3 (Symmetries)**.** *(a) The law of ω is invariant under all graph automorphisms of $\mathbb{Z}^2$.*

(b) If $F_1, F_2 \subseteq E$ have no endpoint in common, then $\omega|_{F_1}$ and $\omega|_{F_2}$ are independent.

(c) $1 - \omega$ has the same law as ω.

Proof. For a finite $F \subseteq E$, $\omega|_F$ is a function of $(S_x)_{x\in V(F)}$, and the S_x are i.i.d. This gives (b). It also gives (a), because the uniform law on 2-subsets of N_x is invariant under the symmetries of $\mathbb{Z}^2$ fixing x. For (c), put $S'_x = N_x \setminus S_x$ if $x_1 + x_2$ is even and $S'_x = S_x$ otherwise. Then (S'_x) has the law of (S_x). Since every edge has exactly one even endpoint, the XOR graph of (S'_x) is $1 - \omega$. □

Proposition 2.4 (No Harris–FKG)**.** *Let e_1, e_2, e_3, e_4 be the edges of a plaquette in cyclic order. Let $\mathcal{A} = \{\omega(e_1) = \omega(e_3) = 1\}$ and $\mathcal{B} = \{\omega(e_2) = 1 \text{ or } \omega(e_4) = 1\}$. Both events are increasing, and $\mathbb{P}(\mathcal{A}\cap\mathcal{B}) - \mathbb{P}(\mathcal{A})\mathbb{P}(\mathcal{B}) = -1/1296$.*

Proof. By Corollary 2.2(b), $\mathbb{P}(\mathcal{A}) = 1/4$ and $\mathbb{P}(\mathcal{B}) = 3/4$. By Corollary 2.2(c), $\mathbb{P}(\mathcal{A}\setminus\mathcal{B}) = \mathbb{P}(\eta = (1,0,1,0)) = \frac{82}{1296}$. Hence $\mathbb{P}(\mathcal{A}\cap\mathcal{B}) = \frac{1}{4} - \frac{82}{1296} = \frac{242}{1296}$, which is less than $\frac{243}{1296} = \mathbb{P}(\mathcal{A})\mathbb{P}(\mathcal{B})$. □

Remark 2.5. The violation is tiny. It is of a specific kind: the model prefers plaquettes with an even number of open edges. All characters are nonnegative, $\mathbb{E}\chi_A \geq 0$, which is a Griffiths-type positivity for the spins $(-1)^{\omega(e)}$. This positivity is not a substitute for Harris–FKG for connection events.

3. Planar duality fails

Let $R_n = \{0, \dots, n+1\} \times \{0, \dots, n\}$, and let H_n be the event that some open path using only edges of R_n joins $\{0\}\times\{0,\dots,n\}$ to $\{n+1\}\times\{0,\dots,n\}$. Let $R_n^* = \{\frac{1}{2}, \dots, n+\frac{1}{2}\}\times\{-\frac{1}{2}, \dots, n+\frac{1}{2}\}$ be the dual rectangle, a copy of R_n rotated by $\pi/2$. By planar duality, H_n fails if and only if R_n^* is crossed from top to bottom by dual edges e^* whose primal edges e are closed. For bond percolation at $p = 1/2$, the closed dual configuration is again bond percolation at $p = 1/2$, hence $\mathbb{P}_{\text{Ber}}(H_n) = 1/2$ for every n.

For the 2-XnG, Proposition 2.3(c) says that the closed edges form a copy of the 2-XnG *as a random subset of E*. The dual crossing, however, uses the adjacency of the dual lattice: two closed edges are consecutive on a dual path if they lie on the boundary of a common face. In the primal, two open edges are consecutive if they share an endpoint. The dependence of the 2-XnG is attached to vertices, so the dependence of the dual configuration is attached to faces, and the dual is a different model.

Proposition 3.1 (The dual is not a copy of the primal)**.** *Let ψ be a graph isomorphism from the dual lattice $(\mathbb{Z}^2)^*$ onto $\mathbb{Z}^2$, for instance $\psi(u) = u - (\frac{1}{2}, \frac{1}{2})$. Let $\tilde{\omega}(\psi(e^*)) = 1 - \omega(e)$ be the closed dual configuration transported to $\mathbb{Z}^2$. For every $x \in V$, with $\mathrm{St}(x) = \{e \in E : x \in e\}$,*

$$\mathbb{E} \prod_{e \in \mathrm{St}(x)} (-1)^{\tilde{\omega}(e)} = \tfrac{1}{81}, \qquad \textit{whereas} \qquad \mathbb{E} \prod_{e \in \mathrm{St}(x)} (-1)^{\omega(e)} = 0.$$

In particular, $\tilde{\omega}$ does not have the law of ω.

Proof. The set $\psi^{-1}(\mathrm{St}(x))$ consists of the four dual edges at the dual vertex $\psi^{-1}(x)$. These are the duals of the four boundary edges of a face p. Hence $\prod_{e \in \mathrm{St}(x)} (-1)^{\tilde{\omega}(e)} = \prod_{f \in \partial p} (-1)^{1-\omega(f)} = \chi_{\partial p}$, which has mean 3^{-4} by Proposition 2.1. The star $\mathrm{St}(x)$ is not even, so $\mathbb{E}\chi_{\mathrm{St}(x)} = 0$. □

Proposition 3.1 does not by itself rule out $\mathbb{P}(H_n) = 1/2$. The following computation does.

Proposition 3.2 (Exact crossing probabilities)**.**

$$\mathbb{P}(H_1) = \tfrac{1}{2}, \qquad \mathbb{P}(H_2) = \tfrac{124345}{248832} \approx 0.4997147, \qquad \mathbb{P}(H_3) = \tfrac{69570879977}{139314069504} \approx 0.4993816.$$

In particular $\mathbb{P}(H_n) < 1/2 = \mathbb{P}_{\mathrm{Ber}}(H_n)$ for $n = 2, 3$.

Proof and computation. The event H_n does not depend on the vertical edges of the left and right sides of R_n. Let F_n be the set of the remaining edges of R_n. By Corollary 2.2(a) and the Walsh expansion of $\mathbf{1}_{H_n}$ on $\{0,1\}^{F_n}$ (Proposition 4.2 below),

$$\mathbb{P}(H_n) = \tfrac{1}{2} + \sum_{\emptyset \neq A \subseteq F_n \text{ even}} 3^{-v_2(A)}\, \hat{h}_n(A), \qquad \hat{h}_n(A) = \mathbb{E}_{\mathrm{Ber}}[\mathbf{1}_{H_n} \chi_A].$$

The even subgraphs of F_n are the sums of the $(n-1)n$ plaquettes of R_n that touch neither side.

- For $n = 1$ there are none, so $\mathbb{P}(H_1) = 1/2$.
- For $n = 2$ there are three nonempty ones. Enumerating $\{0,1\}^{F_2}$ gives $\hat{h}_2 = -47/4096$ for each of the plaquettes $[1,2] \times [0,1]$ and $[1,2] \times [1,2]$, and $\hat{h}_2 = -3/2048$ for the boundary of their union. Hence
$$\mathbb{P}(H_2) = \tfrac{1}{2} + 2 \cdot 3^{-4} \cdot \left(-\tfrac{47}{4096}\right) + 3^{-6} \cdot \left(-\tfrac{3}{2048}\right) = \tfrac{1}{2} - \tfrac{47}{165888} - \tfrac{1}{497664} = \tfrac{124345}{248832}.$$
- For $n = 3$ there are 63 nonempty ones. The value is computed in exact integer arithmetic by enumerating $\{0,1\}^{F_3}$ (2^{25} configurations, `code/fourier3_exact.c`).

Independently, all three values were obtained by a column transfer matrix over the vertex choices in exact integer arithmetic (`code/tm.py`). The state after a column records the connectivity partition of that column, which blocks are joined to the left side, and which column vertices chose their east neighbour. The same program returns exactly $1/2$ for bond percolation ($n = 1, 2$), which tests the geometry. For $n = 2$ the value was also confirmed by enumerating all 6^{12} choice configurations of R_2 (`code/brute_H2.c`). □

Remark 3.3 (A local heuristic)**.** An open primal path can pass straight through a vertex whose other two edges are closed. A closed dual path can pass straight through a face whose other two boundary edges are open. Counting both orientations, the first local pattern has probability $1/8$ (Corollary 2.2(b)) and the second has probability $\frac{1}{8}(1 + \frac{1}{81})$ (Corollary 2.2(c)). So the dual gains

slightly more straight passages than the primal. This is only suggestive. On faces, all patterns with an even number of open edges gain the factor $\frac{82}{81}$ and all odd ones lose $\frac{80}{81}$. The rigorous first-order statement is Proposition 4.3.

4. Vertex-XOR models and the position of the 2-XnG

Definition 4.1. Let ν be a law on subsets of N_o that is invariant under the symmetries of $\mathbb{Z}^2$ fixing o and under complementation $S \mapsto N_o \setminus S$. The *$\nu$-XOR graph* is defined as above, with S_x i.i.d. with law ν (translated to x).

The proof of Proposition 2.1 gives $\mathbb{E}_\nu \chi_A = \prod_x m_\nu(A_x)$ with $m_\nu(B) = \mathbb{E}_\nu(-1)^{|S\cap B|}$. By complement invariance, $m_\nu(B) = (-1)^{|B|} m_\nu(B)$, so $m_\nu(B) = 0$ if $|B|$ is odd. Three cases are of interest.

(i) *Bond percolation at $p = 1/2$.* Take ν uniform on all 16 subsets; equivalently, draw $|S| \sim \mathrm{Bin}(4, \frac{1}{2})$ and then a uniform subset of that size. The half-edge bits $\sigma_x(y)$ are i.i.d. fair, so ω is bond percolation at $p = 1/2$. Its characters are $\mathbf{1}\{A = \emptyset\}$.

(ii) *AB percolation at $p = 1/2$.* Take ν uniform on $\{\emptyset, N_o\}$. Then $\sigma_x \equiv \varepsilon_x$ and $\omega(e) = \varepsilon_x \oplus \varepsilon_y$. The characters are $\mathbf{1}\{A \text{ even}\}$.
- The XOR graph of "straight pairs", ν uniform on $\{\{N, S\}, \{E, W\}\}$, is the same model: with $\tau_x = \mathbf{1}\{S_x = \{N, S\}\}$ one checks $\omega(e) = \tau_x \oplus \tau_y$ for every edge.
- Put $\tilde{\varepsilon}_x = \varepsilon_x \oplus (x_1 + x_2 \bmod 2)$. An edge is open if and only if $\tilde{\varepsilon}$ agrees at its endpoints. So the open clusters are exactly the monochromatic clusters of Bernoulli(1/2) site percolation.
- Since $p_c^{\mathrm{site}}(\mathbb{Z}^2) > 0.556$ [vdBE96] and the transition is sharp [Men86, AB87], $\mathbb{P}_{\mathrm{AB}}(o \leftrightarrow \partial B_n) \le Ce^{-cn}$: AB percolation is strictly subcritical.

(iii) *The 2-XnG.* Take ν uniform on the six 2-subsets. The characters are $3^{-v_2(A)}\mathbf{1}\{A \text{ even}\}$.

Thus critical bond percolation and the 2-XnG differ *only* in the law of the number of chosen neighbours: binomial in the first case, constantly two in the second.

Proposition 4.2 (Expansion around critical bond percolation)**.** *Let $f : \{0,1\}^E \to \mathbb{R}$ depend only on the edges in a finite set F, and let $\hat{f}(A) = \mathbb{E}_{\mathrm{Ber}}[f\chi_A]$, $A \subseteq F$, be its Walsh coefficients. For every ν-XOR model,*

$$\mathbb{E}_\nu f = \sum_{A \subseteq F \text{ even}} \mathbb{E}_\nu[\chi_A]\, \hat{f}(A).$$

In particular,

$$\mathbb{E}_{\mathrm{X}} f = \mathbb{E}_{\mathrm{Ber}} f + \sum_{\emptyset \neq A \subseteq F \text{ even}} 3^{-v_2(A)} \hat{f}(A), \qquad \mathbb{E}_{\mathrm{AB}} f = \mathbb{E}_{\mathrm{Ber}} f + \sum_{\emptyset \neq A \subseteq F \text{ even}} \hat{f}(A).$$

Proof. On $\{0,1\}^F$ we have $f = \sum_{A \subseteq F} \hat{f}(A)\chi_A$, because the χ_A form an orthonormal basis for the uniform measure. Take $\mathbb{E}_\nu$ and use $\mathbb{E}_\nu \chi_A = 0$ for A not even. □

So the 2-XnG arises from the expansion that turns critical bond percolation into the strictly subcritical AB percolation, by damping the term of A by the factor $3^{-v_2(A)} \in (0, 1)$.

The same structure can be phrased as a gauge invariance. Complementing a single S_x flips the four edges at x and preserves every ν-XOR law. Hence ω has the law of $\omega \oplus \delta\varepsilon$, where $(\delta\varepsilon)(\{x, y\}) = \varepsilon_x \oplus \varepsilon_y$ for i.i.d. fair ε independent of ω. In particular, on every box the configuration is uniform given the parities of the numbers of open edges on the plaquettes of the box. These parities are:

- i.i.d. fair for bond percolation;
- all even for AB percolation;

- biased towards even and positively correlated for the 2-XnG, for instance $\mathbb{E}[\chi_{\partial p}\chi_{\partial p'}] = 3^{-6} > 3^{-8}$ for adjacent plaquettes.

The sign of the leading terms can be determined for crossing events. Let p be a plaquette of R_n with corners a, b, c, d in cyclic order. Let E_p be the following event, which concerns only the edges of R_n other than the four edges of ∂p: exactly one corner of p is joined to the left side of R_n, exactly one other corner is joined to the right side, and each of the remaining two corners is joined neither to a side nor to another corner of p.

Proposition 4.3 (Plaquette coefficients of crossing events)**.** *For $h_n = \mathbf{1}_{H_n}$ and every plaquette p of R_n,*

$$\hat{h}_n(\partial p) = -\tfrac{1}{16}\,\mathbb{P}_{\mathrm{Ber}}(E_p) \le 0.$$

Consequently, the single-plaquette part of the expansion of $\mathbb{P}_{\mathrm{X}}(H_n)$ in Proposition 4.2 equals $-\frac{1}{1296}\sum_p \mathbb{P}_{\mathrm{Ber}}(E_p)$.

Proof. Condition on the configuration ζ outside ∂p and put $T(\zeta) = \sum_{\eta\in\{0,1\}^{\partial p}} (-1)^{|\eta|} h_n(\zeta,\eta)$, so that $\hat{h}_n(\partial p) = \frac{1}{16}\mathbb{E}_{\mathrm{Ber}} T$. If the two sides are joined in ζ, then $T = \sum_\eta (-1)^{|\eta|} = 0$. Otherwise T depends only on the partition of $\{a, b, c, d, L, R\}$ induced by ζ, where L and R stand for the left and right sides. An exhaustive check of all such partitions (`code/plaqT.py`) gives the following.

- $T = -1$ for exactly two kinds of partition: (i) one corner lies with L, a different corner lies with R, and the other two corners are singletons; (ii) the blocks of L and R contain one of the diagonals $\{a, c\}$, $\{b, d\}$ each.
- $T = +1$ exactly when the two corners of one diagonal lie with L and with R respectively, and the other diagonal forms a block of its own.
- $T = 0$ in all other cases.

Partitions of type (ii) and those with $T = +1$ are not induced by any configuration. To see this, add two auxiliary vertices ℓ and r in the outer face, joined by open edges to all vertices of the left and of the right side respectively. The resulting graph is still planar, and "joined to the left side" means "joined to ℓ".

- In case (ii), say a, c lie in the cluster of ℓ and b, d in the cluster of r. Choose a self-avoiding open path π from a to c in the first cluster and an open path π' from b to d in the second. Both avoid the edges of ∂p, and they are disjoint. Let $[a, c]$ be the diagonal of p; then $J = \pi \cup [a, c]$ is a Jordan curve. The two open triangles into which $[a, c]$ cuts p lie on different sides of J. Since b and d lie in their closures and not on J, they are separated by J, yet π' joins them without meeting J.
- In the case $T = +1$, say a is joined to ℓ, c to r, and b to d in a third cluster. The same argument applies to $J = \pi_a \cup \gamma \cup \pi_c \cup [c, a]$. Here π_a and π_c are open paths from a to ℓ and from c to r, and γ is an arc from ℓ to r in the outer face.

Hence $T(\zeta) = -\mathbf{1}_{E_p}(\zeta)$ for every ζ. □

Heuristic 4.4. The event E_p is essentially the event that p is pivotal for H_n, so $\sum_p \mathbb{P}_{\mathrm{Ber}}(E_p)$ is of the order of the expected number of pivotal edges. By Russo's formula, this number is the derivative at $1/2$ of the crossing probability of bond percolation with parameter $1/2 + t$. It grows polynomially in n, like $n^{3/4}$ by universality. The first-order correction is therefore $-c\,n^{3/4}/1296$, the same correction one gets for bond percolation with parameter $\frac{1}{2} - t$ for a small $t > 0$.

Larger even subgraphs contribute with both signs. For $n = 3$, 8 of the 63 nonempty even subgraphs of F_3 have positive coefficients, the largest being $96 \cdot 2^{-25}$, and 55 have negative ones, down to $-283350 \cdot 2^{-25}$. In aggregate the higher-order part is negative for $n = 2, 3$ ($-2.0 \cdot 10^{-6}$ and $-7.1 \cdot 10^{-6}$), and the single-plaquette part accounts for 99.3% and 98.8% of the deficit.

Perturbation theory breaks down, and the model shows its subcritical character, once $n^{3/4}/1296$ is of order one. This is what the simulations show.

5. Numerical evidence

All simulations sample the choices (S_x) exactly in finite boxes. Edges inside a box depend only on choices inside it, so the model itself has no boundary effects. The code is in `code/` and the raw output in `data/`. Error bars are ± 2 standard deviations.

5.1. **Crossing probabilities.** Table 1 and Figure 1 show the probability of H_n. Runs with different seeds at the same n are pooled. Control runs for bond percolation at $p = 1/2$ (20,000 samples each) gave 0.5005, 0.4983, 0.5009, 0.4957 and 0.5035 for $n = 8, 16, 32, 64, 128$, all within ± 0.0071 of $1/2$.

A weighted least-squares fit of $\log(\frac{1}{2} - \mathbb{P}(H_n))$ against $\log n$ over $32 \le n \le 1024$ gives slope 0.743. This is the scaling expected for bond percolation slightly off criticality, $\frac{1}{2} - \mathbb{P}_{1/2-t}(H_n) \approx t \cdot \frac{d}{dp}\mathbb{P}_p(H_n) \asymp t\, n^{3/4}$. Over the same range there is no sign that the deficit levels off. At $n = 4096$ the crossing probability is 0.30 ± 0.04.

n	samples	$\mathbb{P}(H_n)$	$\frac{1}{2} - \mathbb{P}(H_n)$
$1, 2, 3$	exact	$0.5,\ 0.4997147,\ 0.4993816$	$0,\ 0.000285,\ 0.000618$
8	20,000	0.5003 ± 0.0071	-0.0003
16	120,000	0.4971 ± 0.0029	0.0029
32	420,000	0.4937 ± 0.0015	0.0063
64	320,000	0.4904 ± 0.0018	0.0096
128	120,000	0.4819 ± 0.0029	0.0181
256	140,000	0.4722 ± 0.0027	0.0278
512	32,000	0.4514 ± 0.0056	0.0486
1024	8,000	0.421 ± 0.011	0.079
2048	3,000	0.363 ± 0.018	0.137
4096	600	0.30 ± 0.04	0.20

Table 1. Probability that $R_n = [0, n+1] \times [0, n]$ is crossed horizontally in the 2-XnG. For bond percolation at $p = 1/2$ it equals $1/2$ for every n.

5.2. **One-arm probabilities.** Table 2 compares $\mathbb{P}(o \leftrightarrow \partial B_n)$ for the 2-XnG and for bond percolation at $p = 1/2$, with 12,000 samples each. For bond percolation, the values for $n = 64$ and $n = 1024$ give an effective exponent of 0.103, close to $5/48 \approx 0.104$, which checks the code. For the 2-XnG the two models agree within error up to $n \approx 256$. After that the 2-XnG falls behind: between $n = 512$ and $n = 1024$ its local exponent is 0.18, against 0.11 for bond percolation.

n	2-XnG	bond percolation, $p = \frac{1}{2}$	ratio
16	0.711	0.709	1.00
64	0.614	0.611	1.00
256	0.519	0.528	0.98
512	0.471	0.494	0.95
1024	0.416	0.459	0.91

Table 2. $\mathbb{P}(o \leftrightarrow \partial B_n)$, 12,000 samples per model; error bars ± 0.009.

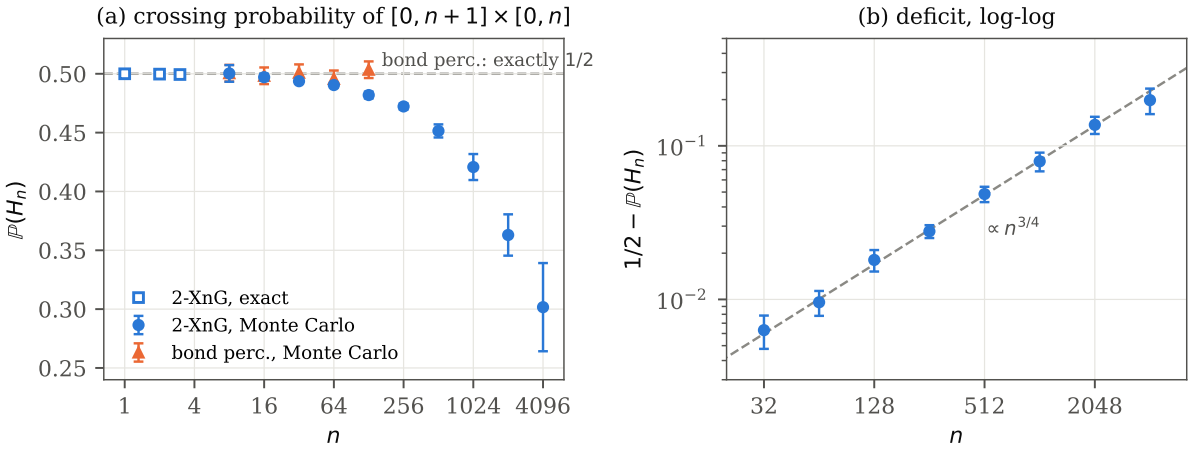


FIGURE 1. (a) $\mathbb{P}(H_n)$ for the 2-XnG (circles: Monte Carlo; squares: exact) and Monte Carlo control runs for bond percolation (triangles). (b) The deficit $\frac{1}{2} - \mathbb{P}(H_n)$ on log-log axes, with a line of slope 3/4.

5.3. **Distance from criticality.** Let $\omega_\delta = \omega \vee \xi_\delta$, where ξ_δ is independent Bernoulli(δ) bond percolation; ω_δ is increasing in δ. Table 3 shows the following.

- At $\delta = 0.0006$ the crossing probabilities still decrease with n.
- At $\delta = 0.0012$ they increase with n.
- At $\delta = 0.0009$ they are constant within error.
- At $\delta = 0.003$ and $n = 1024$ the crossing probability is already 0.678 ± 0.013.

Thus scale invariance is restored at $\delta^* \approx 10^{-3}$, an additional edge density of about $5 \cdot 10^{-4}$. The 2-XnG therefore behaves like bond percolation with parameter about $\frac{1}{2} - 5 \cdot 10^{-4}$. Such a model has a correlation length of order 10^3–10^4, consistent with $\mathbb{P}(H_{4096}) \approx 0.30$.

n	$\delta = 0$	$\delta = 0.0006$	$\delta = 0.0009$	$\delta = 0.0012$
32	0.4937 ± 0.0015	0.4971 ± 0.0016	0.5001 ± 0.0016	0.5017 ± 0.0016
128	0.4819 ± 0.0029	0.4939 ± 0.0041	0.4983 ± 0.0041	0.5042 ± 0.0041
1024	0.421 ± 0.011	0.474 ± 0.014	0.487 ± 0.014	0.531 ± 0.014

TABLE 3. $\mathbb{P}(H_n)$ for $\omega \vee \xi_\delta$ (samples: 400,000, 60,000 and 5,000 per entry for $n = 32, 128, 1024$; the $\delta = 0$ column is Table 1).

6. Consequences for (A) and (B), and what is missing

Lemma 6.1. $\mathbb{P}(o \leftrightarrow \partial B_n) \geq \mathbb{P}(H_n)/(n+1)$. *Hence (B) holds, with $C = 1$, as soon as* $\inf_n \mathbb{P}(H_n) > 0$.

Proof. If H_n occurs, some $(0, j)$ with $0 \leq j \leq n$ is joined to a vertex with first coordinate $n+1$, hence to $\partial B_n((0, j))$. Use a union bound and translation invariance. □

Lemma 6.2 (Finite-size criterion)**.** *Let $q(L) = \mathbb{P}(B_L \leftrightarrow \partial B_{3L}$ inside $B_{3L})$. If $q(L) < 7^{-16}$ for one L, then $\mathbb{P}(o \leftrightarrow \partial B_n) \leq Ce^{-cn}$ for all n, where $c, C > 0$ depend on L and $q(L)$.*

Proof. Call $z \in \mathbb{Z}^2$ *bad* if $2Lz + B_L$ is joined to $\partial(2Lz + B_{3L})$ by an open path inside $2Lz + B_{3L}$. Then $\mathbb{P}(z \text{ bad}) = q(L)$. The boxes $2Lz + B_L$ cover $\mathbb{Z}^2$, and adjacent vertices lie in boxes with $|z - z'|_\infty \le 1$.

Let γ be an open path from o to ∂B_{2LN}. The set of z with $\gamma \cap (2Lz + B_L) \neq \emptyset$ is $*$-connected, contains 0, and reaches $|z|_\infty \ge N$. Every such z with $|z|_\infty \le N-2$ is bad: $2Lz + B_{3L} \subseteq B_{2LN-L}$, so after visiting $2Lz + B_L$ the path γ must leave $2Lz + B_{3L}$. Hence there is a self-avoiding $*$-path of $N-1$ bad sites starting at 0.

A bad event depends only on the choices in $2Lz + B_{3L}$. So bad events of distinct sites in the same residue class modulo 4, which are at ℓ^∞-distance at least 4, are independent. A fixed path of $N-1$ sites has at least $(N-1)/16$ sites in one residue class, and there are at most $8 \cdot 7^{N-3}$ self-avoiding $*$-paths of $N-1$ sites. Therefore

$$\mathbb{P}(o \leftrightarrow \partial B_{2LN}) \le 8 \cdot 7^{N-3}\, q(L)^{(N-1)/16} = \tfrac{8}{49}\big(7\, q(L)^{1/16}\big)^{N-1}. \qquad \square$$

Neither lemma uses FKG or self-duality. Together with the data they show the following.

- (B) holds if $\inf_n \mathbb{P}(H_n) > 0$. In the simulations, $\mathbb{P}(H_n)$ falls from 0.497 to about 0.30 between $n = 16$ and $n = 4096$, and shows no sign of levelling off.
- Exponential decay, which contradicts (B) and implies (A), follows from $q(L) < 7^{-16}$ at a single scale. If the model is subcritical, as the data indicate, this holds for all large L. It cannot be checked rigorously at accessible scales: the correlation length appears to be of order 10^3–10^4, and the threshold is tiny.

Why the standard proofs of (A) do not apply.

(i) Zhang's argument needs a self-dual law, which fails (Section 3). It also needs the square-root trick, which rests on the Harris–FKG inequality, and that inequality is not available (Proposition 2.4).

(ii) The local-to-global principle of [BJKLRT25] does not apply directly. An edge of the XOR graph depends on both endpoints, and conditionally on all other choices the local comparison with bond percolation goes both ways. Let $T_x = \{y : x \in S_y\}$; then the set of open edges at x corresponds to $S_x \triangle T_x$.
- If $T_x = \emptyset$, every set W of at least two edges at x contains an open edge with probability $\ge \frac{5}{6}$, more than under bond percolation.
- If $|T_x| = 2$, no edge at x is open with probability $\frac{1}{6} > \frac{1}{16}$.

(iii) There is no stochastic domination in either direction: the edge marginals agree with Bernoulli(1/2), so domination would force equality of laws.

Possible routes.

(a) *Edge-by-edge interpolation.* Let $\mathbb{P}^{(U)}$ be the model in which the edges of $U \subseteq E$ are replaced by independent fair coins. For f depending on finitely many edges and $a \notin U$, put $\mathrm{Piv}_a(f) = f(\omega^{a,1}) - f(\omega^{a,0})$, where $\omega^{a,i}$ is ω with $\omega(a)$ set to i. The law of $\omega|_{E\setminus\{a\}}$ is the same under $\mathbb{P}^{(U)}$ and $\mathbb{P}^{(U\cup\{a\})}$, so

$$\mathbb{P}^{(U)}(f) - \mathbb{P}^{(U\cup\{a\})}(f) = \mathbb{E}^{(U)}\big[(\omega(a) - \tfrac{1}{2})\,\mathrm{Piv}_a(f)\big].$$

Hence $\mathbb{P}_{\mathrm{X}}(o \leftrightarrow \partial B_n) \le \mathbb{P}_{\mathrm{Ber}}(o \leftrightarrow \partial B_n)$, and with it (A), would follow from the statement: *in the intermediate models, an edge is not more likely to be open when it is pivotal.* Proposition 4.3 is the first-order version of this. Condition on the other three edges of a plaquette having k open edges. The fourth edge is then biased towards open if k is odd and towards closed if k is even. Without conditioning, the odd and even cases balance: $3 + 1$ configurations of the three edges against $1 + 3$. Pivotality excludes $k = 3$, which leaves 3 against 4, a net bias towards closed.

(b) *Fourier signs.* By Proposition 4.2, $\mathbb{P}_{\mathrm{X}}(f) \le \mathbb{P}_{\mathrm{Ber}}(f)$ would follow if $\hat{f}(A) \le 0$ for all nonempty even A. This holds for $f = \mathbf{1}_{H_2}$ but already fails for $\mathbf{1}_{H_3}$ (Heuristic 4.4), although the positive coefficients there are tiny. A proof would have to show that the negative terms dominate.

For one-arm events, the planarity argument of Proposition 4.3 breaks down, because the origin is not in the outer face. We expect positive contributions from eight-arm-type configurations, in which a cluster through two opposite corners of p surrounds the cluster of the origin. These should be of lower order, but this is not proved.

(c) *Finite-size criterion.* A computer-assisted bound $q(L) < 7^{-16}$ is out of reach. A criterion with a much larger threshold, for instance via [LSS97] or a multi-scale argument for crossings of long rectangles, would still require scales beyond the correlation length.

Summary of status (labels as in the project README).

Statement	Status	Evidence
Propositions 2.1–4.3, Lemmas 6.1, 6.2	`unchecked` (proofs given)	reviewed once by a separate AI agent
(A) no percolation	conjecture [JKLT25], strongly supported	Section 5
(B) $\mathbb{P}(o \leftrightarrow \partial B_n) \ge c\, n^{-C}$	very likely `false`	Section 5, Heuristic 4.4
The 2-XnG on $\mathbb{Z}^2$ is subcritical (exponential decay)	new conjecture	Section 5

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